50Days-coding-challenge

Day-16[SQL]

Social Media Data Analysis

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Scenario

You are working as a **Database Analyst** for a new social media platform called **ConnectHub**. The company stores user data, posts, likes, friendships, and comments in a any database. Your task is to **analyze and query the database** to derive insights about user activity and relationships.

The challenge will test your ability to use **subqueries, joins, and stored procedures** in SQL.

Database Schema

1. Users

Column	Type	Description
user_id	INT (PK)	Unique ID of the user
username	VARCHAR	User's profile name
email	VARCHAR	User's email address
join_date	DATE	Date user joined

2. Posts

Column	Type	Description
post_id	INT (PK)	Unique ID of the
post user_id	INT (FK)	Author of the post
content	TEXT	Post text/content
post_date	DATETIME	Date and time of posting

3. Likes

Column Type Description

like_id INT (PK) Unique like ID

Column Type Description

user_id INT (FK) User who

liked post_id INT (FK) Post that was

liked like_date DATETIME Date and time of

like

4. Comments

Column	Type	Description
comment_id	INT (PK)	Unique comment ID
post_id	INT (FK)	Post commented on
user_id	INT (FK)	Comment author
comment_text	TEXT	Comment content
comment_date	DATETIME	Date and time of comment

5. Friendships

Column Type Description

friendship_id INT (PK) Unique ID user_id1

INT (FK) User 1 user_id2 INT (FK)

User 2

since_date DATE Date friendship started

Tasks

Task 1:

1. Retrieve all posts along with the username of the author.

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No limit

Query | Query History

```
select*from users;

SELECT p.post_id, p.content, p.post_date, u.username
FROM Posts p
INNER JOIN Users u ON p.user_id = u.user_id;
```

Data Output | Messages | Notifications

Showing rows: 1 to 40 Page No: 1 of 1

	post_id integer	content text	post_date timestamp without time zone	username character varying (50)
1	1	Control but guy whom sometimes nice pressure get.	2025-06-25 00:41:37	zstokes
2	2	Apply season authority couple writer blue marriage beyond forget first.	2025-04-09 01:52:43	lorianthony
3	3	Human hundred hot race owner member because task company north question natu...	2025-05-21 09:41:05	glenn83
4	4	Should word down avoid onto type true section house life fill pick.	2025-03-19 19:32:52	lorianthony
5	5	Under war check though past herself.	2025-03-20 11:52:40	joel94
6	6	Middle off argue star appear action son every perhaps various TV group ability.	2024-11-30 08:25:37	emily15
7	7	Study finally exactly the baby without.	2024-10-09 13:58:18	jeffrey99
8	8	Factor second very condition decision low about show music entire direction bring.	2025-04-25 12:47:01	mathewschase
9	9	Exactly film contain worry plant tonight wrong paper.	2025-09-15 01:08:57	mathewschase
10	10	Always contain available require box number person focus from already.	2024-11-10 21:34:51	glenn83

Total rows: 40 Query complete 00:00:00.222 CRLF Ln 256, Col 1

2. Find all comments on each post along with the commenter's username.

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No limit

Query | Query History

```
INNER JOIN Users u ON p.user_id = u.user_id;

SELECT c.comment_id, c.post_id, c.comment_text, c.comment_date, u.username
FROM Comments c
INNER JOIN Users u ON c.user_id = u.user_id;
```

Data Output | Messages | Notifications

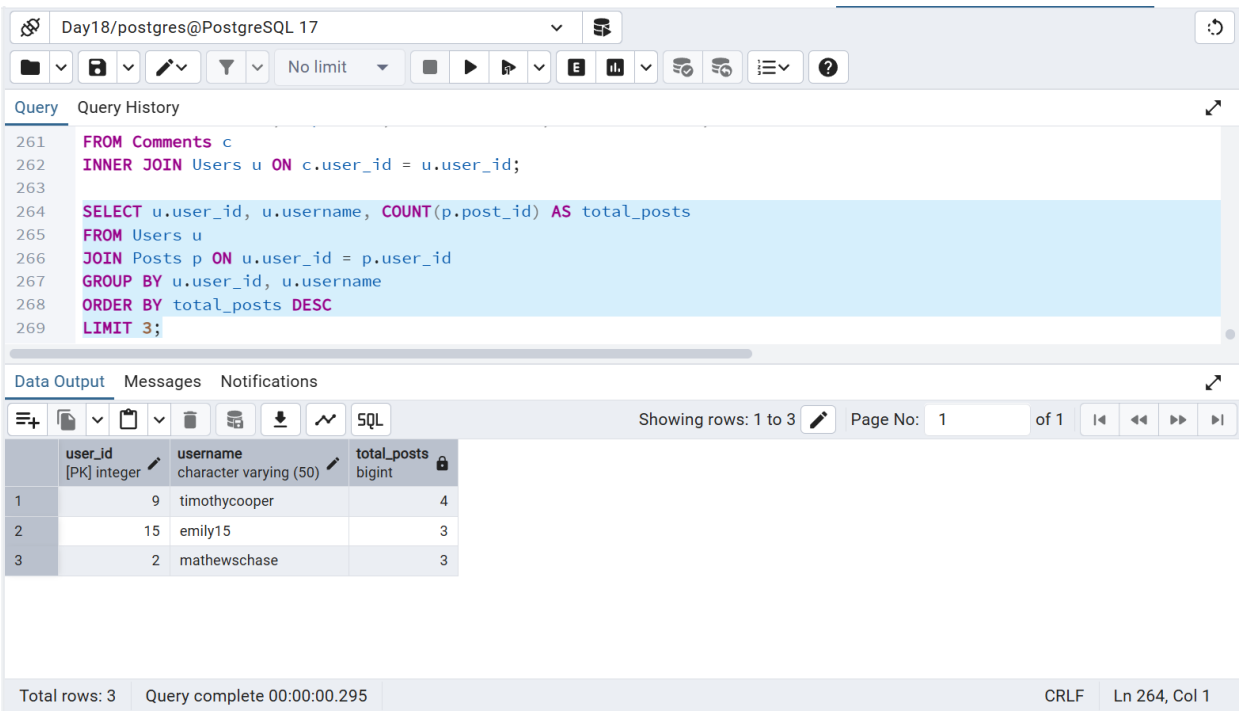
Showing rows: 1 to 50 Page No: 1 of 1

	comment_id integer	post_id integer	comment_text text	comment_date timestamp without time zone	username character varying (50)
1	1	32	Message because protect note fear.	2025-06-08 10:55:35	jeffrey99
2	2	33	Measure none suddenly feeling marriage civil.	2025-03-08 00:29:13	emorse
3	3	30	Learn mind campaign loss pressure management deep act.	2024-12-18 08:00:03	glenn83
4	4	22	Hope tree total chance program past item people.	2025-03-02 17:22:44	brandy78
5	5	21	Know involve agent catch everybody.	2025-02-24 06:08:42	jason52
6	6	19	Player cultural others me possible affect today.	2024-12-18 06:48:44	lorianthony
7	7	40	Experience offer happen rise know determine.	2025-03-01 00:25:37	jason52
8	8	28	Relate until statement rise amount purpose themselves behavior.	2024-10-01 10:12:01	jason52
9	9	17	Become bed occur its.	2024-10-14 15:20:56	lorianthony
10	10	37	Sea south simple moment available case doctor fund.	2025-06-02 01:31:07	emorse

Total rows: 50 Query complete 00:00:00.237 CRLF Ln 260, Col 1

Task 2:

3. Find the top 3 users with the most posts.



	user_id [PK] integer	username character varying (50)	total_posts bigint
1	9	timothycooper	4
2	15	emily15	3
3	2	mathewschase	3

4. Retrieve posts that have more likes than the **average number of likes per post**.

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No limit

Query | Query History

```sql
SELECT p.post_id, COUNT(l.like_id) AS total_likes
FROM Posts p
LEFT JOIN Likes l ON p.post_id = l.post_id
GROUP BY p.post_id
HAVING COUNT(l.like_id) > (
    SELECT AVG(like_count)
```

Data Output | Messages | Notifications

Showing rows: 1 to 9 | Page No: 1 of 1

	post_id [PK] integer	total_likes bigint
1	35	4
2	30	4
3	8	3
4	36	3
5	29	3
6	38	3
7	11	4
8	7	3
9	20	5

Total rows: 9 | Query complete 00:00:00.230 | CRLF | Ln 271, Col 1

5. Find users who have never posted anything but have liked posts.

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No limit

Query | Query History

```sql
    ) AS avg_table
);

SELECT DISTINCT u.user_id, u.username
FROM Users u
JOIN Likes l ON u.user_id = l.user_id
WHERE u.user_id NOT IN (
    SELECT DISTINCT user_id FROM Posts
);
```

Data Output | Messages | Notifications

Showing rows: 1 to 2 | Page No: 1 of 1

	user_id [PK] integer	username character varying (50)
1	18	ajohnson
2	3	bedwards

Total rows: 2 | Query complete 00:00:00.227 | CRLF | Ln 284, Col 1

Task 3:

6. Get a list of all friends of a specific user (say user_id = 3).

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No limit

Query Query History

```
);

SELECT u.user_id, u.username
FROM Users u
WHERE u.user_id IN (
    SELECT user_id1 FROM Friendships WHERE user_id2 = 3
    UNION
    SELECT user_id2 FROM Friendships WHERE user_id1 = 3
);
```

Data Output Messages Notifications

Showing rows: 1 to 2 Page No: 1 of 1

	user_id [PK] integer	username character varying (50)
1	5	mendozacolleen
2	15	emily15

Total rows: 2 Query complete 00:00:00.189 CRLF Ln 290, Col 1

7. Retrieve posts that were liked by **friends of a given user** (nested join scenario).

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No limit

Query Query History

```
SELECT DISTINCT p.post_id, p.content
FROM Posts p
JOIN Likes l ON p.post_id = l.post_id
WHERE l.user_id IN (
    SELECT user_id1 FROM Friendships WHERE user_id2 = 3
    UNION
    SELECT user_id2 FROM Friendships WHERE user_id1 = 3
);
```

Data Output Messages Notifications

Showing rows: 1 to 7 Page No: 1 of 1

	post_id [PK] integer	content text
1	38	Matter hundred store research include form development break democratic artist.
2	23	Church significant happy local whom time lead paper.
3	5	Under war check though past herself.
4	20	Organization call method water serious then surface performance sing.
5	7	Study finally exactly the baby without.
6	30	Why study clearly market girl magazine shoulder rise study through sort southern pull.
7	35	Bring last society somebody participant high financial old all since.

Total rows: 7 Query complete 00:00:00.471 CRLF Ln 298, Col 1

Task 4:

8. Create a stored procedure GetUserActivity(userId) that returns:
    - Total posts by the user
    - Total likes given by the user
    - Total likes received on the user's posts
    - Total comments made by the user

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No limit

Query | Query History

```
        (SELECT COUNT(*)::INT FROM Posts WHERE user_id = userId),
        (SELECT COUNT(*)::INT FROM Likes WHERE user_id = userId),
        (SELECT COUNT(*)::INT FROM Likes
            WHERE post_id IN (SELECT post_id FROM Posts WHERE user_id = userId)),
        (SELECT COUNT(*)::INT FROM Comments WHERE user_id = userId);
END;
$$;

SELECT * FROM GetUserActivity(3);
```

Data Output | Messages | Notifications

Showing rows: 1 to 1 | Page No: 1 of 1

	total_posts integer	total_likes_given integer	total_likes_received integer	total_comments integer
1	0	5	0	2

Total rows: 1 | Query complete 00:00:00.120 | CRLF | Ln 328, Col 1

Task 5: Challenge Query

9. Find the **most influential user** (the user whose posts have the highest total likes + comments).

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No limit

Query | Query History

```sql
SELECT u.user_id, u.username,
       COUNT(DISTINCT l.like_id) + COUNT(DISTINCT c.comment_id) AS influence_score
FROM Users u
JOIN Posts p ON u.user_id = p.user_id
LEFT JOIN Likes l ON p.post_id = l.post_id
LEFT JOIN Comments c ON p.post_id = c.post_id
GROUP BY u.user_id, u.username
ORDER BY influence_score DESC
LIMIT 1;
```

Data Output | Messages | Notifications

Showing rows: 1 to 1 | Page No: 1 of 1

	user_id [PK] integer	username character varying (50)	influence_score bigint
1	9	timothycooper	10

Total rows: 1 | Query complete 00:00:00.276 | CRLF | Ln 341, Col 1

Expected Outcomes

- Demonstrate **understanding of JOINs** (INNER, LEFT, SELF JOIN), **subqueries (scalar, correlated, nested)**, and **stored procedures**.
- They should be able to **interpret real-world problems** into SQL solutions.
- Can optimize queries using **indexes and group by/having clauses**.